FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – VIII

PAPER-1

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**AD**	19.	**ABD**	37.	**AC**
2.	**AD**	20.	**BD**	38.	**AB**
3.	**AC**	21.	**ABCD**	39.	**ABCD**
4.	**AD**	22.	**CD**	40.	**AC**
5.	**ABCD**	23.	**BC**	41.	**AB**
6.	**AB**	24.	**ABC**	42.	**AC**
7.	**BCD**	25.	**ABD**	43.	**ABD**
8.	**B**	26.	**C**	44.	**B**
9.	**D**	27.	**B**	45.	**C**
10.	**C**	28.	**C**	46.	**D**
11.	**C**	29.	**A**	47.	**C**
12.	**B**	30.	**C**	48.	**C**
13.	**A**	31.	**C**	49.	**A**
14.	**3**	32.	**2**	50.	**0**
15.	**1**	33.	**3**	51.	**1**
16.	**4**	34.	**6**	52.	**4**
17.	**2**	35.	**4**	53.	**1**
18.	**2**	36.	**4**	54.	**0**

Physics

PART – I

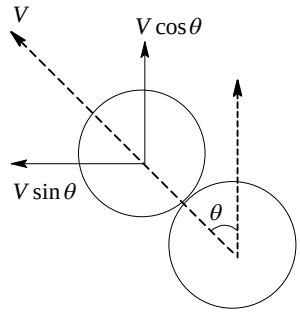
SECTION – A

1. AD

Elastic potential energy $\propto \left(\frac{\partial y}{\partial x}\right)^2$,kinetic energy $\propto \left(\frac{\partial y}{\partial t}\right)^2$. So for antinode elastic potential energy is constant & minimum. $\frac{\partial y}{\partial x}$ always changes for all other points.

2. AD

Let's assume, time taken by the queen to get into the hole is t



$t = \frac{\ell}{2V\sin\theta}$ & also $t = \frac{\ell}{2V\cos\theta} + \frac{\ell}{eV\cos\theta}$

Therefore

$$\frac{\ell}{2V\sin\theta} = \frac{\ell}{2V\cos\theta} + \frac{\ell}{eV\cos\theta}$$

$$\therefore \tan\theta = \frac{1}{3}$$

$$\therefore \frac{d}{\sqrt{(r_1+r_2)^2 - d^2}} = \frac{1}{3}$$

$$d^2 = 9 \Rightarrow d = 3cm$$

3. AC

When parallel light falls from left side

$$\frac{\mu_3}{f} = \frac{(\mu_3 - \mu_2)}{R} + \frac{(\mu_2 - \mu_1)}{R}$$

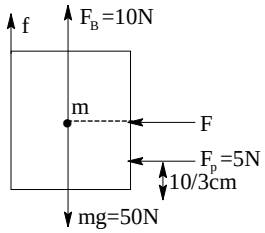
$$\frac{4/3}{f_1} = \frac{(4/3-2)}{R} + \frac{(2-1)}{R} \Rightarrow f_1 = 80cm$$

When parallel light falls from right side

$$\frac{1}{f_2} = \frac{(1-2)}{-R} + \frac{(2-4/3)}{-R} \Rightarrow f_2 = 60cm$$

4. AD

For translatory equilibrium $f + F_B = mg \Rightarrow f = 40N$



For rotational equilibrium $40\times 5 + 5\times\frac{5}{3} = N(5)$

N'=125/3 newton as $F + F_p = N' \Rightarrow F = 110/3\,Newton$

But for $F = 110/3\,Newton$ object will not be in translational equilibrium so for translational equilibrium

$$\frac{9}{10}(F+5) = 40 \Rightarrow F = \frac{355}{9}\text{newton}$$

5. ABCD

6. AB
Normal Reaction on any of the roller always has a horizontal compound directed towards right. Therefore, friction acting on the rollers at every point at contact should have component towards left.
Hence option (C) is incorrect.
Since friction on roller is in opposite direction of F, friction on the plank is in the direction of F. Therefore, for the system to be in equilibrium, F should be less than $mg\sin\theta$ so option A and B are correct.
If rollers are clamped so they cannot move, friction on rollers can be towards left as well as right. Therefore friction on the plank at point A and B can be opposite to F, hence F can be greater then $mg\sin\theta$, so option D is incorrect.

7. BCD
$$0.85=\frac{13.6Z^2}{n_1^2} and\ 0.544=\frac{13.6Z^2}{n_2^2}$$
As $n_2-n_1=3\Rightarrow n_1=12, n_2=15$
$$\frac{hc}{\lambda}=RhcZ^2\left(\frac{1}{12^2}-\frac{1}{15^2}\right)\Rightarrow\lambda=4052nm$$

8. B

9. D

10. C

11. C

12. B

13. A

Sol. **[Q. 11 – 13]**
Use $\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

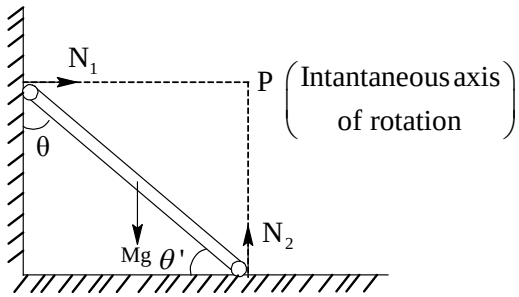
SECTION – C

14. 3
x coordinate of centre of mass of the rod
$$x=\frac{\ell}{2}\sin\theta$$
$$v_x=\frac{\ell}{2}\cos\theta\omega\ and\ a_x=\frac{\ell}{2}\left(\cos\theta\alpha-\sin\theta\omega^2\right)$$
When $N_1=0\Rightarrow\cos\theta\alpha=\sin\theta\omega^2---(1)$
Using COME:
$$mg\frac{\ell}{2}(\cos\beta-\cos\theta)=\frac{1}{2}m\frac{\ell^2}{4}\omega^2+\frac{1}{2}m\frac{\ell^2}{12}\omega^2$$

$\left[v = \frac{\ell}{2}\omega \text{ from IAOR} \right]$

$\omega^2 = \frac{3g}{\ell}(\cos\beta - \cos\theta) ---(2)$

$\alpha = \frac{3g}{2\ell}\sin\theta ----(3)$

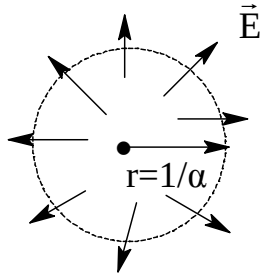


Substituting the value of ω^2 and α in equation (1) gives

$\Rightarrow \cos\theta = \frac{2}{3}\cos\beta \Rightarrow \cos\theta = \frac{1}{\sqrt{3}} \Rightarrow \sin\theta' = \frac{1}{\sqrt{3}}$ so N=3

15. 1

As the electric field is radial, by applying gauss law, we can write



$\int \vec{E}.d\vec{s} = \frac{Q}{\varepsilon_0}$

For $r = \frac{1}{\alpha}$, $\vec{E} = C\left(1 - e^{-\alpha \times 1/\alpha}\right)\frac{\hat{r}}{(1/\alpha)^2}$

$\therefore \oint \vec{E}.d\vec{s} = C\left(1 - e^{-1}\right)\alpha^2 \times 4\pi (1/\alpha)^2$

$\Rightarrow \frac{Q}{\varepsilon_0} = 4\pi C\left(1 - e^{-1}\right) \Rightarrow Q = \left(1 - e^{-1}\right) \Rightarrow \therefore N = 1$

16. 4

$\int_0^{x_{\min}} 0.5x\frac{mg}{\ell}dx = 2 \Rightarrow x_{\min} = 40\text{cm}$

17. 2

$mr^2\beta = \int_0^{\theta} \frac{2q}{2\pi} d\alpha\, vBr\cos\alpha$

$\beta = \frac{dw}{dt} = \frac{qvB\sin\theta}{\pi\ mr} ----(1)$

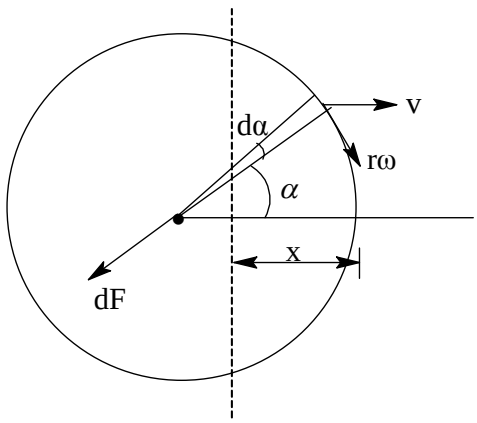
$-ma = \int_0^{\theta} 2\frac{q}{2\pi} d\alpha\, rwB\cos\alpha$

$a = \frac{dv}{dt} = \frac{qrwB\sin\theta}{\pi\ m} ----(2)$

$= \frac{dw}{dv} = \frac{v}{r^2 w} \Rightarrow r^2 \int_0^w w dw = -\int_{v_0}^{v} v\, dv$

$\Rightarrow v = \frac{v_0}{\sqrt{2}}$

$\frac{vdv}{dx} = \frac{qrwB\sin\theta}{\pi m}$



x=r-rcosθ

dx=rsinθdθ

$$\int_{v_0}^{v_0/\sqrt{2}} \frac{vdv}{\sqrt{v_0^2 - v^2}} = \frac{qrB}{m\pi}\int_0^{\pi} \sin^2\theta d\theta$$

$$\Rightarrow q = \frac{\sqrt{2}mv_0}{Br}$$

18. 2

For slight displacement $\left(\pi r^2 x\right)\rho g = ma \Rightarrow T = 2\pi\sqrt{\frac{m}{\pi r^2 \rho g}} \Rightarrow T = 2\,\text{sec}$

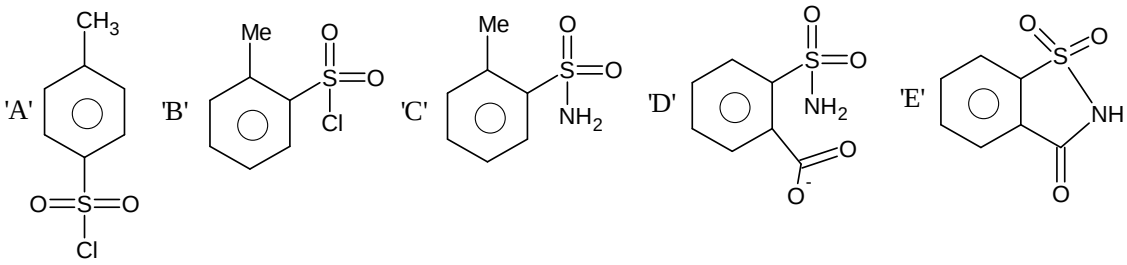
Chemistry

PART – II

SECTION – A

19. ABD

Sol.



20. BD

A.O. is a single e^- wave function

Area of the plot for r<52.9 pm is smaller than the area of the plot of r>52.9pm.

At higher atomic number every of $3d<4s$

$$\Delta E_{2\to5} = RhcZ^2\left[\frac{1}{2^2}-\frac{1}{5^2}\right] = RhcZ^2\left[\frac{21}{100}\right]$$

$$\Delta E_{2\to4} = RhcZ^2\left[\frac{1}{2^2}-\frac{1}{4^2}\right] = RhcZ^2\times\frac{3}{16}$$

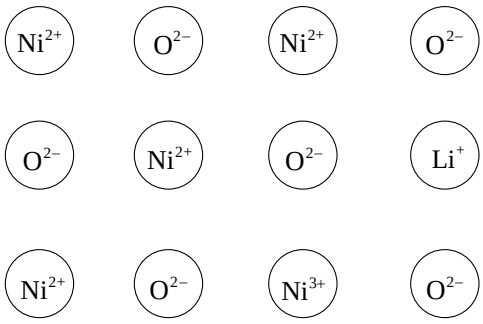
$$\Delta E_{2\to5} = \frac{28}{25}\Delta E_{2\to4}$$

21. ABCD

For dissociative process (e.g.H_2 to 2H), more pressure is required to get the same extent of chemisorption as two species have to be chemisorbed.

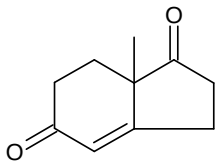
In anionic surfactant negatively charged $-COO^-$ group will repeal each other so at lower concentration micelles will form.

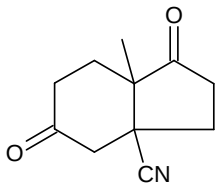
22. CD



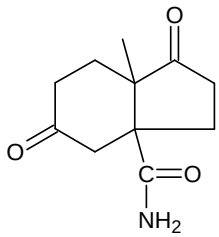
Li^+ ion occupy Ni^{2+} sites to form substitutional defects. In order to maintain the charge neutrality, every Li^+ ion is balanced by Ni^{3+} ion and it becomes a p-type semiconductor.

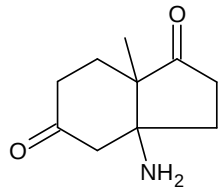
23. BC

'A' 

'B' 

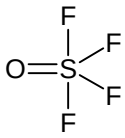
$(1,4\ addition)$

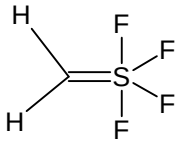
'C' 

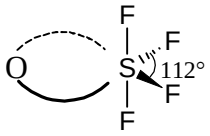
'D' 

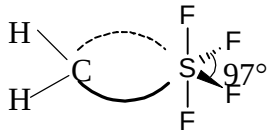
24. ABC

F-F < Cl-Cl < Br-Br < I-I (Bond length)



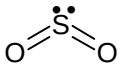


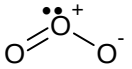




Oxygen being more EN, the electron density at S in OSF_4 will decreases so it occupies less space so bond angle $F_{eq.}$ - S - $F_{eq.}$ will increased.

$H_2S(92°) < O_3(116.8°) < SO_2(119.5°) < NO_2(134°)$





$AsH_3 < NH_3 < SbH_3 < H_2O(BP)$

25. ABD

Rate of reaction = $k_2[Cl][CHCl_3]$ and $\frac{k_1}{k_{-1}} = \frac{[Cl]^2}{[Cl_2]}$

$$[Cl] = \sqrt{\frac{k_1}{k_{-1}}[Cl_2]}$$

Rate of reaction = $k_2 . \sqrt{\frac{k_1}{k_{-1}}}[Cl_2]^{1/2}[CHCl_3]$

$$k = 1.3\times10^{-2} \times \sqrt{\frac{4.8\times10^3}{1.2\times10^3}} = 2.6\times10^{-2}$$

26. C
27. B
28. C

Sol. **(Q. 26 – 28)**

(I) Isothermal expansion

(II) Water & Ice will remain in equilibrium to 0°C & 1 atm pressure so $\Delta G = 0$

(III) For adiabatic reversible $T_2P_2^{\frac{1-\gamma}{\gamma}} = T_1P_1^{\frac{1-\gamma}{\gamma}} \Rightarrow T_2 = 300\left(\frac{P_1}{P_2}\right)^{-2/5} = 300\left(\frac{10}{1}\right)^{-2/5} = 120K$

For adiabatic irreversible

$$C_v(T_2-T_1) = -P_{ext}(V_2-V_1) = -P_2\left[\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right]$$

$$\frac{C_v}{nR}(T_2-T_1) = -P_2\left[\frac{T_2}{P_2} - \frac{T_1}{P_1}\right]$$

$$\frac{(T_2-T_1)}{\gamma-1} = -T_2 + \frac{T_1P_2}{P_1}$$

$$T_2 = T_1\left[\frac{(\gamma-1)}{\gamma}\frac{P_2}{P_1} + \frac{1}{\gamma}\right]$$

$$T_2 = 192K$$

(IV) Mixing of ideal gas at constant T & P in an isolated container

$\Rightarrow q = 0, w = 0, \Delta U = 0, \Delta S_{system} > 0, \Delta G < 0$

29. A
30. C
31. C

Sol. **(Q. 29 to 31)**

Increasing in polarity of the solvent will increase the rate if T.S. has more charge density in comparison to reactant**.**

SECTION – C

32. 2

$$\% \text{ of carbon} = \frac{12\times 2.9\times 100}{44} = 79.09$$

$$R\text{-}OH + Na \longrightarrow R\text{-}ONa + \frac{1}{2}H_2\,(0.00275\,\text{mole})$$

Moles of oxygen $= 0.00275\times 2$

Mass of oxygen $= 0.00275\times 2\times 16 = 0.088g$

% of oxygen = 8.8

So % hydrogen $= 100 - 79.09 - 8.8 = 12.11$

Empirical = $C_{12}H_{22}O$

So $\frac{x+y}{17} = \frac{12+22}{17} = 2$

33. 3

In equilibrium, vapour pressure of solution will remain same

$$\frac{\Delta p}{p^*} = \left(\frac{n_{sugar}}{n_{sugar}+n_{H_2O}}\right)_{\text{Beaker B}} = \left[\frac{2\ n_{NaCl}}{2\ n_{NaCl}+n_{H_2O}}\right]_{\text{BeakerA}}$$

$$\frac{0.1}{0.1+\frac{(1-a)\times 1000}{18}}=\frac{2\times 0.1}{2\times 0.1+\frac{(1+a)\times 1000}{18}}$$

$\Rightarrow a=1/3$

So molarity of NaCl $=\frac{0.1}{\frac{4}{3}}=\frac{0.3}{4}=\frac{3}{40}$

34. 6

I,II,III,IV,V,VI shows cis & trans isomerism

35. 4

Sol. Pressure of O_2 gas filled inside bulb $=(76+76)\,cm\,Hg=2\,atm$

O_2 is a limiting reagent.

$$CH_4+2O_2\rightarrow CO_2+2H_2O_{(l)}$$

4 mole 2 mole

$PV=nRT$

$2\times V=(4)(0.08)(280)$

$V=44.8L=11.2\times 4L$

$x=4$

36. 4

$Fe^{2+},Zn^{2+},Ni^{2+},Cu^{2+}$

Forms precipitate as FeS,ZnS,NiS,CuS respectively Al^{3+},Cr^{3+} ions ppt as hydroxides

Fe^{3+} forms precipitate as $FeS(\text{not } Fe_2S_3)$.

Mathematics

PART – III

SECTION – A

37. AC

$$I_{n+k}=\int_{\frac{n+k}{2}}^{\frac{n+k+1}{2}}\frac{\sin\left(\pi\sin^2\pi x\right)}{\left(\sqrt{2}\right)^x}dx$$

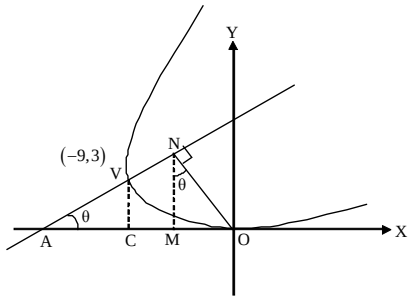
Let $x=\frac{k}{2}+t$

$$I_{n+k}=\int_{\frac{n}{2}}^{\frac{n+1}{2}}\frac{\sin\left(\pi\sin^2\pi\left(\frac{k}{2}+t\right)\right)}{\left(\sqrt{2}\right)^{\frac{k}{2}+t}}dt$$

$$I_{n+k}=\frac{1}{2^{\frac{k}{4}}}\int_{\frac{n}{2}}^{\frac{n+1}{2}}\frac{\sin\left(\pi\sin^2\pi t\right)}{\left(\sqrt{2}\right)^t}dt$$

$$I_{n+k}=\frac{I_n}{2^{\frac{k}{4}}}$$

38. AB



$VC=3units, CO=9units$

As per property: $AV=VN$ *and* $\frac{AN}{AV}=2=\frac{MN}{VC}$

$\Rightarrow OM=6\tan\theta, AC=CM=3\cot\theta\Rightarrow 9=3\left(2\tan\theta+\frac{1}{\tan\theta}\right)$

$\Rightarrow m=1 or\ m=\frac{1}{2}$

$\Rightarrow Eq^n$ of Axis can be

$(y-3)=1(x+9)$

or

$(y-3)=\frac{1}{2}(x+9)$

39. ABCD

Let the series be $21, 21r, 21r^2, ...$

$\text{Sum} = \dfrac{21}{1-r}$ is a positive integer

Also 21r is a positive integer

$S = \dfrac{(21)(21)}{21-21r}$ as $21r \in N$ hence 21-21 r must be an integer

Also $21r < 21$

Hence 21-21r may be equal to 1, 3, 7 or 9

i.e must be a divisor of (21) (21)

$\text{hence } 21-21r = 1 \text{ or } 3 \text{ or } 7 \text{ or } 9$

21r=20, 18, 14 or 12

40. AC

$f(x)$ is continuous on a closed interval so it attains a minimum value α.

Since α is in the range of f, $\therefore f(\alpha) = 1$. If $\alpha = 1, f(x) = 1 \forall x \text{ and } \int_0^1 f(x)\,dx = 1$

Now, if $\alpha < 1$, by intermediate value theorem, since f is continuous it attains all values between $\alpha \text{ and } 1$. So for all $x \geq \alpha, f(x) = 1$.

There fore

$$\int_0^1 f(x)\,dx = \int_0^\alpha f(x)\,dx + (1-\alpha)$$

Since $f(x) \geq \alpha, \int_0^\alpha f(x)\,dx > \alpha^2$ and the equality is strict because f is continuous and thus cannot be α for all $x < \alpha \text{ and } 1 \text{ at } \alpha$. So

$$\int_0^1 f(x)\,dx > \alpha^2 + (1-\alpha) = \alpha\left[\alpha - \frac{1}{2}\right]^2 + \frac{3}{4} \geq \frac{3}{4}$$

$$\therefore \frac{3}{4} < \int_0^1 f(x)\,dx \leq 1$$

41. AB

$$T = |\vec{a}||\vec{b}||\vec{c}||\cos\alpha||\sin\theta_1|$$

$$= |\vec{a}||\vec{b}||\vec{c}||\cos\alpha||\sin\theta_2|$$

$$= |\vec{a}||\vec{b}||\vec{c}||\cos\alpha||\sin\theta_3|$$

$$Y = 2\left(|\vec{a}\times\vec{b}| + |\vec{b}\times\vec{c}| + |\vec{c}\times\vec{a}|\right)$$

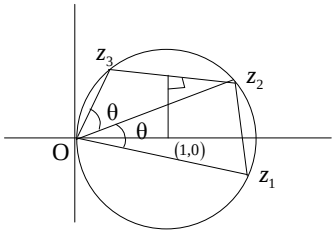
$$= 2\left(|\vec{a}||\vec{b}||\sin\theta_1| + |\vec{b}||\vec{c}||\sin\theta_2| + |\vec{c}||\vec{a}||\sin\theta_3|\right)$$

$$\frac{Y}{T} = \frac{2}{|\cos\alpha|}\left(\frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|}\right) = 4\left(\frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|}\right) \Rightarrow |\cos\alpha| = \frac{1}{2}$$

42. AC

$\lim_{x\to 0} f(x)\, f(2x)$ is -1 or 0 depending upon $x = \frac{1}{2^n}$ or $x \neq \frac{1}{2^n}$ but $f(x) + f(2x)$ always tends towards zero $f(2) = f\left(\frac{1}{2^{-1}}\right) = -1$ but $\lim_{x\to 2} f(x)$ is 0, $\lim_{x\to 3} f(x) = f(3) = 0$

43. ABD



1st option – chord $(Z_3 - Z_2)$ is $\perp$ to line joining $\frac{z_2 + z_3}{2}$ and 1.

2nd option – angle by chord $(Z_3 - Z_2)$ at $(1,0)$ is double of angle at $(0,0)$

4th option – Ptolemy's theorem

44. B

For option B total cost of all the n trips= $\frac{n\lambda}{2}(d_{PA} + d_{PB})$

i.e. we have to minimise $d_{PA} + d_{PB}$.

$d_{PA} + d_{PB} \geq d_{AB}$ where equality holds when p lies on line segment joining A & B

45. C

For option C total cost of all the n trips

$= \frac{n\lambda}{3}(d_{PA} + d_{PB} + d_{PC})$

$d_{PA} + d_{PB} + d_{PC}$ is minimum when P is interior point such that AB, BC, CA subtends $120°$ at P.

46. D

For option D total cost of all the n trips

$= \frac{n\lambda}{3}(d_{PA} + d_{PB} + 2d_{PC})$

As $d_{PA} + d_{PC} \geq d_{AC}$ & $d_{PB} + d_{PC} \geq d_{BC}$

$\Rightarrow d_{PA} + d_{PB} + 2d_{PC} \geq d_{AC} + d_{BC}$

Equality holds when P is vertex C

47. C

$A^3 = B^3$ & $A^2B = B^2A$ subtracting both we get

$A^2(A - B) = B^2(B - A)$

$\Rightarrow (A^2 + B^2)(A - B) = 0$

$\Rightarrow \det.(A^2 + B^2) = 0$ (otherwise $A = B$ which is not true)

48. C

49. A

Sol. **(Q. 48 to 49)** By given information

$a_{ij}=\begin{cases}2 & \text{when } i=j\\ 0 & \text{when } i\neq j\end{cases}$

$A=\left[a_{ij}\right]$

$A=\begin{bmatrix}2 & 0 & 0 & 0\\ 0 & 2 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 0 & 2\end{bmatrix}$

$|A|=2^4$

$|\text{adj}\,A|=|A|^{n-1}=(16)^3=2^{12}$

$\Rightarrow |\text{adj}(\text{adj}\,A)|=|\text{adj}\,A|^3=2^{36}=\left(2^3\right)^{12}=(1+7)^{12}$

$={}^{12}C_0+{}^{12}C_1\,7^1+....+{}^{12}C_{12}\;7^{12}$

$\Rightarrow \left\{\frac{2^{36}}{7}\right\}=\frac{1}{7}$

Given $A=(I+B)(I-B)^{-1}$

Now, $AA^T=(I+B)(I-B)^{-1}(I+B)^{-1}(I-B)$

$=(I+B)(I+B)^{-1}(I-B)^{-1}(I-B)=I\times I=I$

$\therefore |AA^T|=|I|\Rightarrow\ |A|^2=1\Rightarrow |A|=\pm 1$

Given $|A|>0\therefore |A|=1$

$\therefore \det(2A)-\det(\text{adj}\,A)$

$=2^3\det A-(\det A)^2$

$=8\times 1-1^2=7$

$a_{ij}=\begin{cases}I_{6+i,3}-I_{i+3,3}, & i=j\\ 0, & i\neq j\end{cases}$

$I_{6+i,3}-I_{3+i,3}=\int_0^1\left(\frac{x^{i+6}}{x^3-1}-\frac{x^{i+3}}{x^3-1}\right)dx=\int_0^1 x^{i+3}dx=\frac{1}{i+4}$

$\Rightarrow A=\begin{bmatrix}\frac{1}{5} & 0 & 0\\ 0 & \frac{1}{6} & 0\\ 0 & 0 & \frac{1}{7}\end{bmatrix}\Rightarrow |A|=\frac{1}{5\times 6\times 7}=\frac{1}{210}$

For $B=\begin{bmatrix} J_{6,5} & 72 & J_{11,5} \\ J_{7,5} & 63 & J_{12,5} \\ J_{8,5} & 56 & J_{13,5} \end{bmatrix}$

Applying $C_1 \to C_1+C_3$

$$|B|=\begin{vmatrix} J_{6,5}+J_{11,5} & 72 & J_{11,5} \\ J_{7,5}+J_{12,5} & 63 & J_{12,5} \\ J_{8,5}+J_{13,5} & 56 & J_{13,5} \end{vmatrix}$$

$$J_{6,5}+J_{11,5}=\int_0^1 \frac{x^6}{x^5+1}dx+\int_0^1 \frac{x^{11}}{x^5+1}dx=\int_0^1\left(x-\frac{x}{x^5+1}\right)dx+\int_0^1\left(x^6-x+\frac{x}{x^5+1}\right)dx=\frac{1}{7}$$

$$\begin{vmatrix} \frac{1}{7} & 72 & J_{11,5} \\ \frac{1}{8} & 63 & J_{12,5} \\ \frac{1}{9} & 56 & J_{13,5} \end{vmatrix}=\frac{1}{7\times 8\times 9}\begin{vmatrix} 72 & 72 & J_{11,5} \\ 63 & 63 & J_{12,5} \\ 56 & 56 & J_{13,5} \end{vmatrix}=0$$

SECTION – C

50. 0

The required line shall be represented by

$(y-x)+\lambda_1(z-1)=0 \;\&\; (y+x)+\lambda_2(z+1)=0 \ldots\ldots\ldots\ldots(1)$

Where (x,y,z) is any general point on the line.

At $z=0, x^2-y^2=1$

$(y-x)=\lambda_1 \ldots\ldots\ldots\ldots\ldots(2)$

$(y+x)=-\lambda_2 \ldots\ldots\ldots\ldots\ldots(3)$

$(2)\times(3)$

$\Rightarrow y^2-x^2=-\lambda_1\lambda_2$

$\Rightarrow -1=-\lambda_1\lambda_2 \Rightarrow \lambda_1\lambda_2=1$

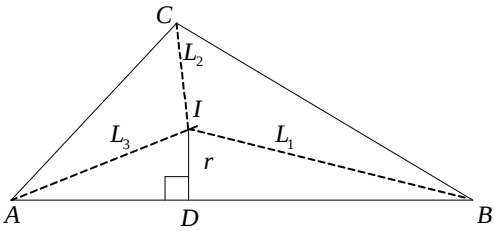
Substituting $\lambda_1 \;\&\; \lambda_2$ from (1)

$$\frac{(y-x)}{(z-1)}\times\frac{y+x}{z+1}=1\Rightarrow y^2-x^2=z^2-1$$

$x^2-y^2+z^2-1=0$

$\alpha+\beta+\gamma+\delta=0$

51. 1



Intersection point of $x-2y+6=0 \, and \, x-y+2=0$ is the Incentre of the ΔABC

$I(2,4)$

$r =$ in radius = Perpendicular distance of I from the line AB

$$= \left| \frac{2+12-4}{\sqrt{10}} \right|$$

$$r = \sqrt{10}$$

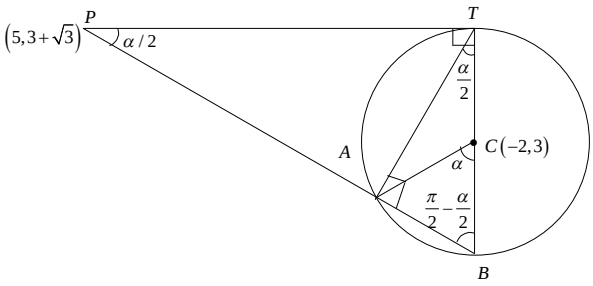
$$\sum \left(r \cot \frac{A}{2} + a \right) = r \cot \frac{A}{2} + a + r \cot \frac{B}{2} + b + r \cot \frac{C}{2} + c$$

$$= s - a + a + s - b + b + s - c + c$$

$$= 3s$$

$$\left[\frac{\Delta}{\sum \left(r \cot \frac{A}{2} + a \right)} \right] = \left[\frac{rs}{3s} \right] = \left[\frac{\sqrt{10}}{3} \right] = 1$$

52. 4



$CT = CB = r = 4$

$$PT = \sqrt{5^2 + \left(3+\sqrt{3}\right)^2 + 4 \times 5 - 6\left(3+\sqrt{3}\right) - 3}$$

$PT = 6$

Let $\angle ACB = \alpha$

$$Ar\left(\Delta CAB\right) = \frac{1}{2} \cdot r \cdot r \cdot \sin\alpha = 8\sin\alpha$$

$$Ar\left(\Delta CAT\right) = \frac{1}{2} \cdot r \cdot r \cdot \sin\left(\pi - \alpha\right) = 8\sin\alpha$$

In ΔPBT

$\tan\frac{\alpha}{2}=\frac{BT}{PT}=\frac{8}{6}$

$\Rightarrow \sin\alpha=\frac{2\tan\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}}=\frac{2\times\frac{4}{3}}{1+\frac{16}{9}}=\frac{24}{25}$

$Ar\left(\Delta CAB\right)+Ar\left(\Delta CAT\right)=16\sin\alpha=16\times\frac{24}{25}=\frac{384}{25}$

$\Rightarrow \lambda=384$ hence $\left[\sqrt{384}\right]-15=19-15=4$

53. 1

Satisfying then given solution in differential equation

get $\left(\lambda_1\left(a^2-2a+1\right)+\lambda_2\left(\left(b^2-2b+1\right)x+2\left(b-1\right)\right)\right)e^{ax}=0$ which must be true for every $\lambda_1 \& \lambda_2 \in R$ so $a^2-2a+1=0, b^2-2b+1=0 \Rightarrow a=b=1$

54. 0

$(1,2)\ and\ (3,6)$ are foci of ellipse

$2ae=2\sqrt{5}\Rightarrow 2a=\frac{2\sqrt{5}}{e}=2+\sqrt{5}+\sqrt{45}$

$\sqrt{\left(\sin\theta-1\right)^2+\left(\cos\theta-2\right)^2}+\sqrt{\left(\sin\theta-3\right)^2+\left(\cos\theta-6\right)^2}=\left(1+\sqrt{45}\right)+\left(1+\sqrt{5}\right)$

$\left(1+\sqrt{5}\right)and\left(1+\sqrt{45}\right)$ are maximum distance of $(1,2)$ are $(3,6)$ from circle

$x^2+y^2=1\Rightarrow (1,2),(3,6),(\alpha,\beta)$ are collinear

$2\alpha-\beta=0$